

390 Digital Technology Concepts (Open)

(formerly 391 Information Technology Concepts)

Description & Eligibility

Contestants will demonstrate foundational knowledge of the information technology field, including core concepts across hardware, software, operating systems, networking, security, and technology applications. Any secondary or postsecondary division contestant may enter this objective event.

Contestant Must Supply

- (Optional) Cordless calculator: Electronic devices will be monitored according to ACT standards. See NLC Calculator Guidelines. Contestants who violate this rule will be disqualified

Contest Notes

- Method of evaluation: objective
- AI Policy Level 1 *No AI Allowed*
- Reference materials are not allowed. Contestants who violate this rule will be disqualified.
- Length of event: no more than sixty (60) minutes testing time
- Unlimited entries by state

Contest Competencies

1. Demonstrate foundational understanding of core IT concepts, including hardware, software, operating systems, and networking, and apply practical skills to configure, manage, and troubleshoot these components for secure and efficient operations
2. Explain the evolution of programming languages and apply this knowledge to evaluate language paradigms, features, and appropriate use cases when selecting technologies for various projects
3. Demonstrate understanding of cybersecurity principles, including risk management, access control, authentication, threat awareness, and best practices for protecting information systems
4. Demonstrate understanding of legal, ethical, and policy considerations in IT environments, including data privacy, intellectual property, compliance standards, and responsible technology use
5. Demonstrate foundational knowledge of database concepts, including relational structures, SQL basics, data integrity, storage practices, and the role of data management in organizational operations
6. Demonstrate understanding of operating system concepts such as file systems, process management, user permissions, configuration tools, and system utilities as they relate to supporting business technology environments
7. Demonstrate foundational knowledge of emerging and modern IT technologies—including cloud computing, virtualization, IoT, automation, and AI-related concepts—and evaluate their impact on organizational efficiency and technology strategy